**Requirement Traceability Matrix – TrustLoker**

## 1. Kamus domain

Kode D-01 sampai D-21 mengikuti urutan kelompok pada class diagram dan dipakai sebagai kolom pada bagian 4.

| Kode | Kelas | Peran dalam sistem |
|---|---|---|
| D-01 | User (abstrak) | Akun dasar: email, kata sandi, OTP, kunci login, token |
| D-02 | Student | Mahasiswa: cari, lamar, laporkan, simpan favorit |
| D-03 | CompanyUser | PIC perusahaan: kelola listing, tanggapi laporan, kelola pelamar |
| D-04 | Admin | Peninjau verifikasi, laporan, sanggahan, rule, dan akun |
| D-05 | Notification | Notifikasi dan email otomatis beserta status kirim |
| D-06 | AuditLog | Catatan tindakan admin dan perubahan konfigurasi |
| D-07 | Company | Profil, legalitas, status verifikasi dan status akun perusahaan |
| D-08 | RegistrationCheck | Hasil pengecekan domain, website, dan kontak saat registrasi |
| D-09 | TrustScore | Skor mentah, skor tampil 0–100, dan badge |
| D-10 | ScoreHistory | Riwayat nilai lama, nilai baru, penyebab, dan waktu |
| D-11 | VerificationSubmission | Pengajuan verifikasi (badge) beserta dokumen dan keputusan |
| D-12 | JobListing | Lowongan beserta status dan alasan peringatan |
| D-13 | ListingHistory | Riwayat perubahan listing |
| D-14 | Application | Lamaran mahasiswa: CV, pesan, persetujuan, status |
| D-15 | ListingActivity | Aktivitas mahasiswa pada listing: favorit dan riwayat dilihat |
| D-16 | Report | Laporan komunitas beserta bukti, status, dan keputusan |

| Kode | Kelas | Peran dalam sistem |
|---|---|---|
| D-17 | RightOfReply | Sanggahan perusahaan dalam jendela 3x24 jam |
| D-18 | Sanction | Sanksi: lowongan dicabut atau perusahaan ditangguhkan |
| D-19 | DetectionRule | Rule deteksi: kata kunci, pola, ambang look-alike |
| D-20 | DetectionResult | Hasil deteksi rule dan domain pada listing |
| D-21 | «service» ScheduledJob | Job harian dan bulanan: kedaluwarsa, poin, retensi |

**2. Ketertelusuran FR, use case, dan kelas**

| FR | Requirement | Prioritas | Use case | Kelas terkait |
|---|---|---|---|---|
| FR-STU-01 | Registrasi mahasiswa dengan OTP dan consent | Must | UC-01 | User, Student, Notification |
| FR-STU-02 | Login, logout, reset password | Must | UC-01 | User |
| FR-STU-03 | Cari dan filter lowongan | Must | UC-02 | Student, JobListing, Company, TrustScore |
| FR-STU-04 | Detail lowongan, badge, skor, riwayat verifikasi | Must | UC-03 | Student, JobListing, Company, TrustScore, VerificationSubmission, ListingActivity |
| FR-STU-05 | Laporkan lowongan atau perusahaan | Must | UC-04 | Student, Report, JobListing, Company, Notification |
| FR-STU-06 | Favorit dan riwayat lowongan dilihat | Should | UC-05 | Student, ListingActivity, JobListing |
| FR-STU-07 | Pantau status laporan | Should | UC-06 | Student, Report, Notification |
| FR-STU-08 | Kelola profil dan hapus akun | Should | UC-05 | User, Student, Report |
| FR-STU-09 | Melamar lowongan dengan CV | Must | UC-23 | Student, Application, JobListing, Notification |
| FR-STU-10 | Pantau status lamaran | Should | UC-23 | Student, Application, Notification |

| FR | Requirement | Prioritas | Use case | Kelas terkait |
|---|---|---|---|---|
| FR-COM-01 | Registrasi perusahaan dengan OTP | Must | UC-07 | User, CompanyUser, Company, RegistrationCheck, Notification |
| FR-COM-02 | Ajukan verifikasi (badge) | Must | UC-08 | CompanyUser, Company, VerificationSubmission |
| FR-COM-03 | Kelola listing lowongan | Must | UC-09 (include UC-18) | CompanyUser, JobListing, ListingHistory, DetectionResult |
| FR-COM-04 | Terima laporan dan ajukan right of reply | Must | UC-10 | CompanyUser, Report, RightOfReply, JobListing, Notification |
| FR-COM-05 | Dashboard ringkas perusahaan | Should | UC-11 | CompanyUser, Company, JobListing, TrustScore, Report |
| FR-COM-06 | Kelola profil perusahaan | Should | UC-11 | CompanyUser, Company, VerificationSubmission |
| FR-COM-07 | Lihat dan kelola pelamar | Must | UC-24 | CompanyUser, Application, JobListing, Company, Notification |
| FR-ADM-01 | Tinjau pengajuan verifikasi | Must | UC-12 (include UC-19) | Admin, VerificationSubmission, Company, TrustScore, AuditLog, Notification |
| FR-ADM-02 | Triase laporan komunitas | Must | UC-13 | Admin, Report, Notification |
| FR-ADM-03 | Kelola rule deteksi | Should | UC-15 | Admin, DetectionRule, AuditLog |
| FR-ADM-04 | Dashboard analitik | Could | UC-17 | Admin, Report, TrustScore, Company |
| FR-ADM-05 | Putuskan sanggahan dan hasil laporan | Must | UC-14 (include UC-19) | Admin, Report, RightOfReply, Sanction, TrustScore, Notification |
| FR-ADM-06 | Tangguhkan atau pulihkan akun dan listing | Must | UC-16 | Admin, Company, JobListing, Sanction, AuditLog |

| FR | Requirement | Prioritas | Use case | Kelas terkait |
|---|---|---|---|---|
| FR-ADM-07 | Lihat dan telusuri audit log | Should | UC-17 | Admin, AuditLog |
| FR-ADM-08 | Tinjau lowongan yang ditandai otomatis | Must | UC-22 | Admin, JobListing, DetectionResult, RightOfReply, Sanction, AuditLog |
| FR-SYS-01 | Deteksi otomatis rule-based | Must | UC-18 | DetectionRule, DetectionResult, JobListing, Company, Notification |
| FR-SYS-02 | Deteksi domain look-alike | Must | UC-18 | DetectionRule, DetectionResult, Company |
| FR-SYS-03 | Hitung dan perbarui trust score | Must | UC-19 | TrustScore, ScoreHistory, Company |
| FR-SYS-04 | Notifikasi email otomatis | Should | UC-20 | Notification, User |
| FR-SYS-05 | Penjadwalan harian | Must | UC-21 | ScheduledJob, JobListing, TrustScore, ScoreHistory, Company, Notification |
| FR-SYS-06 | Audit log otomatis | Must | UC-17 | AuditLog, Admin |
| FR-SYS-07 | Penghapusan atau anonimisasi data retensi | Should | UC-21 | ScheduledJob, Report, Application |
| FR-SYS-08 | Pengecekan otomatis saat registrasi perusahaan | Must | UC-07 | RegistrationCheck, Company |

**3. Matriks silang FR dan kelas domain**

**Bagian A: D-01 sampai D-11**

| FR | D-01 | D-02 | D-03 | D-04 | D-05 | D-06 | D-07 | D-08 | D-09 | D-10 | D-11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| FR-STU-01 | × | × | | | × | | | | | | |
| FR-STU-02 | × | | | | | | | | | | |
| FR-STU-03 | | × | | | | | × | | × | | |
| FR-STU-04 | | × | | | | | × | | × | | × |

| FR | D-01 | D-02 | D-03 | D-04 | D-05 | D-06 | D-07 | D-08 | D-09 | D-10 | D-11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| FR-STU-05 | | × | | | × | | × | | | | |
| FR-STU-06 | | × | | | | | | | | | |
| FR-STU-07 | | × | | | × | | | | | | |
| FR-STU-08 | × | × | | | | | | | | | |
| FR-STU-09 | | × | | | × | | | | | | |
| FR-STU-10 | | × | | | × | | | | | | |
| FR-COM-01 | × | | × | | × | | × | × | | | |
| FR-COM-02 | | | × | | | | × | | | | × |
| FR-COM-03 | | | × | | | | | | | | |
| FR-COM-04 | | | × | | × | | | | | | |
| FR-COM-05 | | | × | | | | × | | × | | |
| FR-COM-06 | | | × | | | | × | | | | × |
| FR-COM-07 | | | × | | × | | × | | | | |
| FR-ADM-01 | | | | × | × | × | × | | × | | × |
| FR-ADM-02 | | | | × | × | | | | | | |
| FR-ADM-03 | | | | × | | × | | | | | |
| FR-ADM-04 | | | | × | | | × | | × | | |
| FR-ADM-05 | | | | × | × | | | | × | | |

| FR | D-01 | D-02 | D-03 | D-04 | D-05 | D-06 | D-07 | D-08 | D-09 | D-10 | D-11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| FR-ADM-06 | | | | × | | × | × | | | | |
| FR-ADM-07 | | | | × | | × | | | | | |
| FR-ADM-08 | | | | × | | × | | | | | |
| FR-SYS-01 | | | | | × | | × | | | | |
| FR-SYS-02 | | | | | | | × | | | | |
| FR-SYS-03 | | | | | | | × | | × | × | |
| FR-SYS-04 | × | | | | × | | | | | | |
| FR-SYS-05 | | | | | × | | × | | × | × | |
| FR-SYS-06 | | | | × | | × | | | | | |
| FR-SYS-07 | | | | | | | | | | | |
| FR-SYS-08 | | | | | | | × | × | | | |
| Jumlah FR | 5 | 9 | 7 | 9 | 14 | 6 | 16 | 2 | 8 | 2 | 4 |

**Bagian B: D-12 sampai D-21**

| FR | D-12 | D-13 | D-14 | D-15 | D-16 | D-17 | D-18 | D-19 | D-20 | D-21 |
|---|---|---|---|---|---|---|---|---|---|---|
| FR-STU-01 | | | | | | | | | | |
| FR-STU-02 | | | | | | | | | | |
| FR-STU-03 | × | | | | | | | | | |
| FR-STU-04 | × | | | × | | | | | | |
| FR-STU-05 | × | | | | × | | | | | |
| FR-STU-06 | × | | | × | | | | | | |
| FR-STU-07 | | | | | × | | | | | |
| FR-STU-08 | | | | | × | | | | | |
| FR-STU-09 | × | | × | | | | | | | |

| FR | D-12 | D-13 | D-14 | D-15 | D-16 | D-17 | D-18 | D-19 | D-20 | D-21 |
|---|---|---|---|---|---|---|---|---|---|---|
| FR-STU-10 | | | × | | | | | | | |
| FR-COM-01 | | | | | | | | | | |
| FR-COM-02 | | | | | | | | | | |
| FR-COM-03 | × | × | | | | | | | × | |
| FR-COM-04 | × | | | | × | × | | | | |
| FR-COM-05 | × | | | | × | | | | | |
| FR-COM-06 | | | | | | | | | | |
| FR-COM-07 | × | | × | | | | | | | |
| FR-ADM-01 | | | | | | | | | | |
| FR-ADM-02 | | | | | × | | | | | |
| FR-ADM-03 | | | | | | | | × | | |
| FR-ADM-04 | | | | | × | | | | | |
| FR-ADM-05 | | | | | × | × | × | | | |
| FR-ADM-06 | × | | | | | | × | | | |
| FR-ADM-07 | | | | | | | | | | |
| FR-ADM-08 | × | | | | | × | × | | × | |
| FR-SYS-01 | × | | | | | | | × | × | |
| FR-SYS-02 | | | | | | | | × | × | |
| FR-SYS-03 | | | | | | | | | | |
| FR-SYS-04 | | | | | | | | | | |
| FR-SYS-05 | × | | | | | | | | | × |
| FR-SYS-06 | | | | | | | | | | |
| FR-SYS-07 | | | × | | × | | | | | × |
| FR-SYS-08 | | | | | | | | | | |
| Jumlah FR | 13 | 1 | 4 | 2 | 9 | 3 | 3 | 3 | 4 | 2 |